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GREEN'S FUNCTION AND SUPER-PARTICLE METHODS FOR KINETIC SIMULATION OF HETEROEPITAXY

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Arrays of nanosized three dimensional islands are known to self-assemble spontaneously on strained heteroepitaxial thin films. We simulate the dynamics using kinetic Monte Carlo method based on a ball and spring lattice model. Green's function and super-particle methods which greatly enhance the computational efficiency are explained.

Keywords: Strained heteroepitaxy; kinetic Monte Carlo; Green's function.

1. Green's Function Approach for Elastic Energy Calculation

Dislocation-free epitaxial Ge films deposited on Si(100) substrates are compressively strained due to the 4% misfit in the lattice constants between Ge and Si. At less than 3 monolayers, a flat wetting layer of Ge covers the Si surface due to its lower surface energy. Upon further deposition, three dimensional (3D) Ge islands form spontaneously as an effective way to relief the elastic stress.[1] They are in the form of pyramids bounded by four (105) facet planes. They then further develop into dome islands bounded mainly by (113) facets.[2] Domes has a higher aspect ratio and relief the stress even better. Only if deposition continues will dislocated islands appear. Similar self-assembled 3D islands are also observed in other film-substrate combinations such as InAs/GaAs and InAs/InP as well as their alloy generalizations including Ge_xSi_{1-x}/Si and $In_xGa_{1-x}/GaAs$.[3] These 3D islands are of much current research interest since they behave as quantum dots with unique electronic properties. They are expected to find applications in future microelectronic devices.

Computer modeling has provided important insights for a range of thin film deposition problems. Simulation of the morphological evolution of strained films is however much more challenging due to the long-range nature of the elastic interactions. An effective approach is kinetic Monte Carlo (KMC) method based on square or cubic lattices of balls and springs representing atoms and elastic interactions in two dimensional (2D)[4,5,6] or 3D[7,8] simulations respectively. Atoms that are nearest or next nearest neighbors are directly connected. This defines the elastic properties of the system. Further, a simple chemical bond energy term depending on the coordination numbers of the atoms is also present. Each surface atom m can hop randomly to a site at a neighboring column with a probability proportional to $\exp[-(\kappa_m - \Delta E_m)/kT]$, where κ_m is the bond energy of atom m and ΔE_m is the change of the elastic energy E_s of the system upon removal of the atom. Physically, a weakly bonded or highly stressed atom is more likely to hop. By repeatedly simulating the hopping of atoms according to the above probability, morphological evolution of the film can be simulated. The computation is very intensive because formation of an island involves a large number of atomic hopping events and hence many repeated calculations of the elastic energy.

Large scale KMC simulation have become possible using improved algorithms including a Green's function approach which dramatically speeds up the elastic energy calculations.[5,7] Using 3D simulation, interesting dependence of the morphology on temperature has been demonstrated. Figure 1 shows the results from annealing of initially flat films 10 MLs thick at 6% misfit.[7] The substrate used contains $64 \times 64 \times 64$ atoms with a lattice constant 2.72 Å which gives an atomic density the same as that of Si. At 1000 K [Figure 1(a)], closely packed islands are obtained. However, at 600 K [Figure 1(b)], sharp grooves which develops from 3D pits (not shown) are obtained. Similar islands, pits and grooves have also been observed in experiments. The roughening mechanisms in the two regimes are indeed different and are expected to be described by the Asaro-Tiller-Grinfeld surface instability and a layer-by-layer nucleation theory respectively.[7]

Each simulation requires of the order of 10^6 calculations of the elastic energy E_s of the spring system which dominate the program run time. In principle, each calculation requires the solution of the equilibrium positions of all atoms and involves solving a huge set of linear equations. In practice, our Green's function method and a super-particle approach lead to dramatically reduced sets of equations and this will be explained in detail in the following.

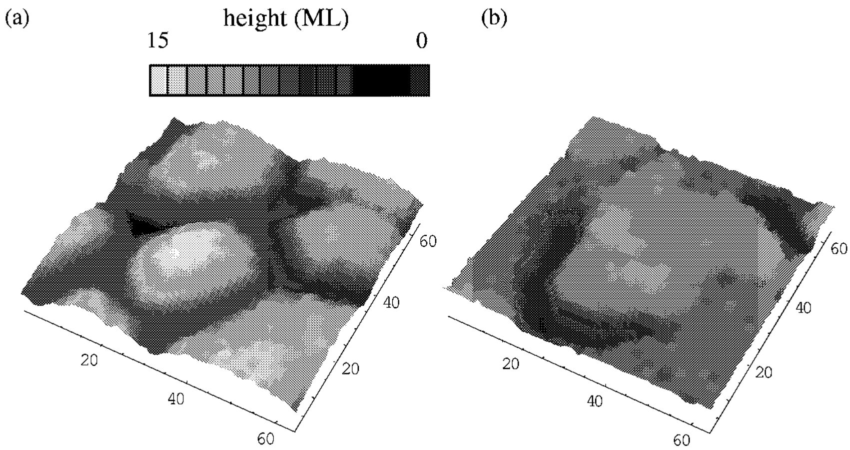

Fig. 1. Surfaces from simulation of annealing of initially flat films of 10 ML with 6% misfit at (a) 1000 K and time 100μs, and (b) 600 K and time 0.25 s.

We first define a displacement $\vec{u}_i$ due to the elastic strain for each atom i measured from a reference lattice position corresponding to a flat homogenously strained heteroepitaxial film. From Hook's law and after linearization, the elastic force $\vec{f}_{ij}$ on atom i follows

$$\vec{f}_{ij} = -\mathbf{K}_{ij}(\vec{u}_i - \vec{u}_j) + \vec{b}_{ij} \tag{1}$$

where the 3×3 matrix $\mathbf{K}_{ij}$ is an elastic modulus tensor and the constant vector $\vec{b}_{ij}$ essentially represents the stress due to the lattice mismatch. The net force on each atom in the film and the substrate vanishes at equilibrium and in principle leads to a huge set of coupled equations from which $\vec{u}_i$ can be obtained. Our Green's function approach is a lattice analog of boundary integral methods. The idea is to remove the explicit reference to all bulk atoms under the film surface from the coupled equations. This is done by representing these indirect long-range interactions using appropriate lattice Green's functions. Specifically, we extend the lattice to attain a flat upper boundary by filling extra atoms on top of the surface with identical elastic properties to those of the film. We call these "ghost" atoms. The springs connecting the real and ghost atoms are unphysical. However, we can fix this by applying additional external forces to exactly eliminate the effects of the ghost atoms. The displacement of any surface atom i can then be shown to follow[5]

$$\vec{u}_i = \sum_{jj'} [(\mathbf{G}_{ij} - \mathbf{G}_{ij'})\mathbf{K}_{jj'}\vec{u}_j - \mathbf{G}_{ij}\vec{b}_{jj'}] \tag{2}$$

The summation is over all real and ghost surface atoms j and j' respectively which are directly connected neighbors. The solution of equation (2) is already a much faster method to obtain the displacement $\vec{u}_i$ of every surface atom. The elastic energy E_s then follows easily by applying the principle of virtual work.

We have been able to further improve the computational performance by an additional super-particle (SP) coarsening approach. Since finding the elastic energy ΔE_m of atom m requires calculating E_s of the whole lattice twice with and without the atom. Obviously some surface details far away from the hopping site will not contribute much to ΔE_m and can be neglected. Specifically, the surface atoms are divided into groups. Each group is called a super-particle. We assume that all atoms in a SP has *identical displacement* $\vec{u}_i \equiv \vec{u}_I$ for each of its member $i \in \Omega_I$, where Ω_I denotes the set of atoms constituting super-particle I. The linear size of a SP is proportional to its distance from the hopping site. Figure 2 shows a schematic diagram of the partition of all 64×64 columns into SPs. The hopping site is located at a random one of the 4 columns at the center and note that periodic boundary conditions are assumed. Each surface atom that is within the central region of 8×8 columns is simply treated as a trivial SP and contain only one single atom. Outside the 8×8 region there are square shells of SPs. The first, second and third shells have SPs each of size 2×2, 4×4 and 8×8 respectively.

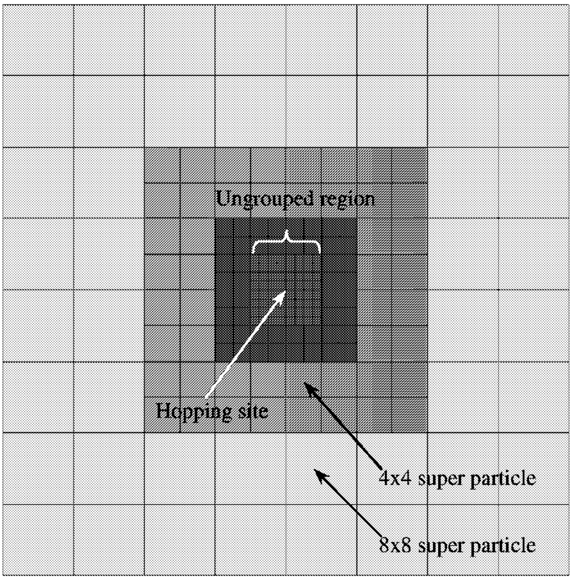

Fig. 2. Partitioning of a 64×64 surface into super-particles. The central 8×8 region contains trivial super-particles. Shells of super-particles are shaded in different gray levels.

Equation (2) is then approximated by

$$\vec{u}_I = \sum_J \left\{ \left[\sum_{j\in\Omega_J} \sum_{j'} (\mathbf{G}_{Ij} - \mathbf{G}_{Ij'})\mathbf{K}_{jj'} \right] \vec{u}_J - \sum_{j\in\Omega_J} \sum_{j'} \mathbf{G}_{Ij} \vec{b}_{jj'} \right\} \tag{3}$$

where $\mathbf{G}_{Ij} = \mathbf{G}_{ij}$ with the lattice point i at the centroid of the set. By solving equation (3) with a bi-conjugate gradient method, we can get the displacement $\vec{u}_i$ of the surface atoms and hence the elastic energy E_s of the system. Using this approach, the number of unknown $\vec{u}_i$ can be reduced from about 5000 to about 200 for a 64×64 substrate. The number of unknowns will scale up roughly as $\log(L)$ for a $L \times L$ substrate. However, the operations needed to calculate individual coefficients in equations are still enormous for large SPs. To handle this problem, we first rewrite equation (3) as

$$\vec{u}_I = \sum_J \left\{ \left[\sum_{j\in\Omega_J} \sum_{j'} \delta_{jS} \Delta\mathbf{G}_{Ij\beta} \mathbf{K}_{j\beta} \right] \vec{u}_J - \sum_{j\in\Omega_J} \sum_{S} \delta_{jS} \mathbf{G}_{Ij} \vec{b}_{j\beta} \right\} \tag{4}$$

where the Sth bond of atom j connects to atom β. Moreover, $\Delta\mathbf{G}_{Ij\beta} = \mathbf{G}_{Ij} - \mathbf{G})_{I\beta}$ and

$$\delta_{jS} = \begin{cases} 1 & \text{if } S\text{th bond of atom } j \text{ is broken} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

Now we can further approximate equation (4) by

$$\vec{u}_I = \sum_J \left\{ \left[\sum_S n_{JS} \Delta \mathbf{G}_{IJ\beta} \mathbf{K}_{J\beta} \right] \vec{u}_J - \sum_S n_{JS} \mathbf{G}_{IJ} \vec{b}_{J\beta} \right\} \tag{6}$$

where

$$n_{JS} = \sum_{j \in \Omega_J} \delta_{jS} \tag{7}$$

represents the number of broken bonds in SP J in direction S. Now the SPs are completely characterized by their centroid positions indexed by I and J.

Equation (6) provides a good approximation of the coupling coefficients only when the SPs I and J are far apart. We still apply the more accurate equation (3) instead for SPs that are close to each other. In our simulation, the error due to the super-particle approximation is typically within 5 percent as shown in Fig. 3. We have also performed small scale simulations confirming that our super-particle approximation leads to no noticeable difference in the final film morphology.

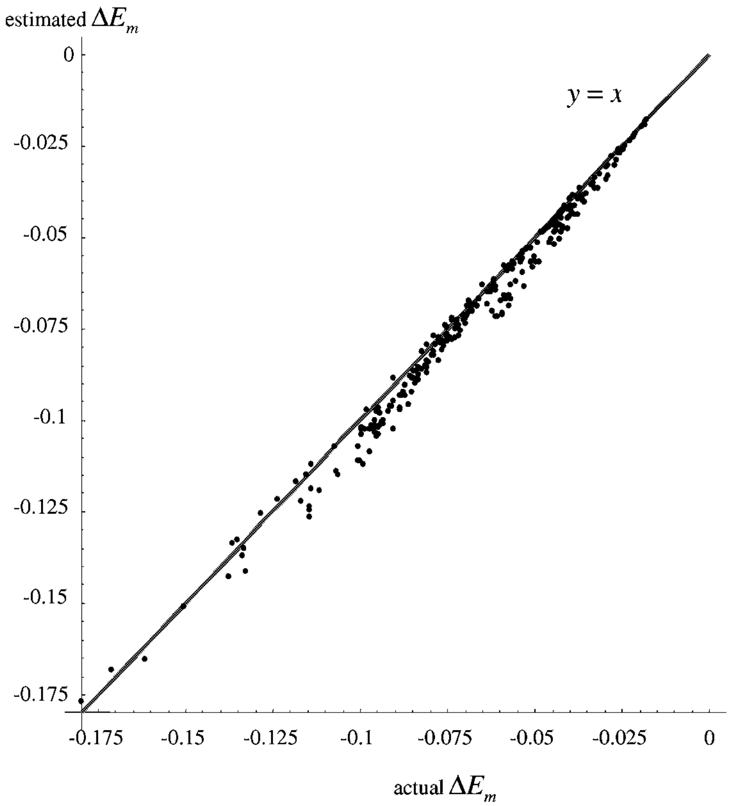

Fig. 3. Plot of estimated ΔE_m using super-particle method against actual ΔE_m.

In conclusion, roughening of strained heteroepitaxy can be studied using kinetic Monte Carlo method in 3D. Accelerated algorithms including Green's function method and superparticle approach greatly improve the computational performance of the simulations.

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